IoT-Based Predictive Temperature Analysis: A Study on effects of Humidity, Light Intensity and UV level on Temperature in early year conditions at the NIC canteen dining area

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Advanced Diploma in Data Science 24.1F/CO

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Date of Submission – 27/01/2025

A Project Proposal submitted for the partial fulfillment of the requirements of the Advanced Diploma in Data Science (Full time) Programme

Students' Declaration

I hereby declare that the work presented in this project report was carried out independently by myself and have cited the work of others and given due reference diligently.

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17/03/2025

Acknowledgement

We express our sincere gratitude to all those who contributed to making this research and our project a success. Our most heartfelt appreciation directs towards our supervisor, Ms. W.M.S.G.D.C Wanigasekara, for her guidance, insights and support during this study. We would also like to extend our appreciation towards our lecturers Mr. D.D.A. Arthanayake and Ms. J.B. DonPaul for their immeasurable guidance.

We are also thankful to the many faculty staff at National Innovation Center for their resource provision and encouragement. Special thanks go to the NIC management staff, NIC canteen management, Janitorial staff and security staff for allowing and extending their help for us to conduct research within the premises and for their cooperation in data collection.

Finally many thanks go to our friends and relatives who inspired and motivated us through this journey.

Executive Summary

This research explores the relationship between ambient environmental factors and temperature variations in institutional dining spaces, with a specific focus on the NIC canteen in Colombo, Sri Lanka. Given the dynamic occupancy levels and external climatic influences in such settings, understanding how variables like humidity, UV intensity, and light levels interact with temperature is crucial for enhancing thermal comfort and energy efficiency.

The study integrates quantitative data collected through IoT-based environmental monitoring with statistical modelling which involves thorough data exploration and creation on a multiple linear regression model, and measuring it's competency through performance metrics, aiming to develop a predictive framework for temperature fluctuations.

The research is expected to contribute to data-driven environmental management strategies, supporting more efficient decision-making in institutional dining areas. By generating new knowledge in predictive temperature regulation, this study provides a foundation for future applications in sustainability and indoor and outdoor climate optimization.

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Chapter 1: Introduction

1.1 Background

Institutional dining areas, such as university canteens, experience continuous environmental fluctuations due to varying occupancy levels and external climatic conditions. However, the impact of ambient environmental factors such as humidity, UV intensity, and light levels on temperature variations remains underexplored in these dynamic settings. While IoT-based environmental monitoring has gained traction in smart building automation, much of the existing research focuses on residential and office environments rather than high-occupancy institutional spaces. Similarly, predictive modeling techniques, particularly multiple linear regression, have been widely used for temperature forecasting, yet they often rely on historical climate data rather than real-time sensor inputs. By integrating real-time IoT sensor data with predictive temperature modeling, this study aims to provide valuable insights into the relationship between ambient environmental factors and indoor temperature dynamics. The findings can contribute to more effective thermal comfort management and potential energy efficiency improvements in similar high-traffic indoor environments.

1.2 Research Problem

Institutional dining areas, such as the NIC canteen, experience dynamic environment fluctuations due to varying occupancy and external climatic conditions which can cause thermal discomfort for occupants, affecting their dining experience and overall well being.

Despite the growing adoption of IoT-based environmental monitoring (Polymeni et al. 2022), there is a lack of comprehensive analysis on how temperature variations correlate with key ambient factors such as humidity, UV intensity, and light levels in institutional dining settings. Non data driven approaches to climate control are still utilized among many institutional spaces including the NIC canteen, and this gap highlights the need for a predictive framework integrating statistical modeling techniques to enhance thermal comfort by making data driven approaches to climate control.

1.3 Research Questions

The main focus of this study in order to address the problem stated in section 1.2 is given as follows:

Can an IoT based predictive model improve temperature forecasting in institutional dining spaces using multiple ambient environmental factors such as humidity, UV intensity and light level?

1.4 Objective of the Project

The main objective of this research and analysis is to identify whether a statistically significant relationship between humidity, UV intensity and light level with temperature exists.

1.5 Scope of the Project

According to the above mentioned details in sections 1.2, 1.3 and 1.4, the scope of this research is the analysis of ambient environmental factors with temperature within the NIC canteen in early year environmental conditions.

1.6 Justification of the Research

An IoT monitoring and analysis system ensures a low cost method of sampling a large set of data points, which leads to accurate predictions (Cochran 1977; Polymeni et al. 2022). This research is scalable in many fields including sustainable temperature regulation systems in outdoor environments and data driven climate prediction systems. The basic system proposed by this research also allows temperature monitoring and analysis which could aid in making data-driven decisions.

1.7 Limitations

This research shows a limited scope of generalization due to the study being conducted on a single point in one institutional dining area, which disregards other environments such as office spaces or residential buildings.

Since the data collection period is limited to the early year time frame, an absence of data from seasonal variations is perceived which ignores climate trends (Werndl 2016).

The research is only limited to four environmental factors which disregard other factors which could potentially influence the response variable (Cochran 1977).

Hardware errors due to calibration issues, measurement errors or physical obstructions as well as network disruptions or power failures which may delete data can produce incomplete data sets which affect model reliability (Zhuang et al. 2023).

The research assumes linear relationships among variables which may be inaccurate in more complex environments (Cochran 1977).

1.8 Work schedule

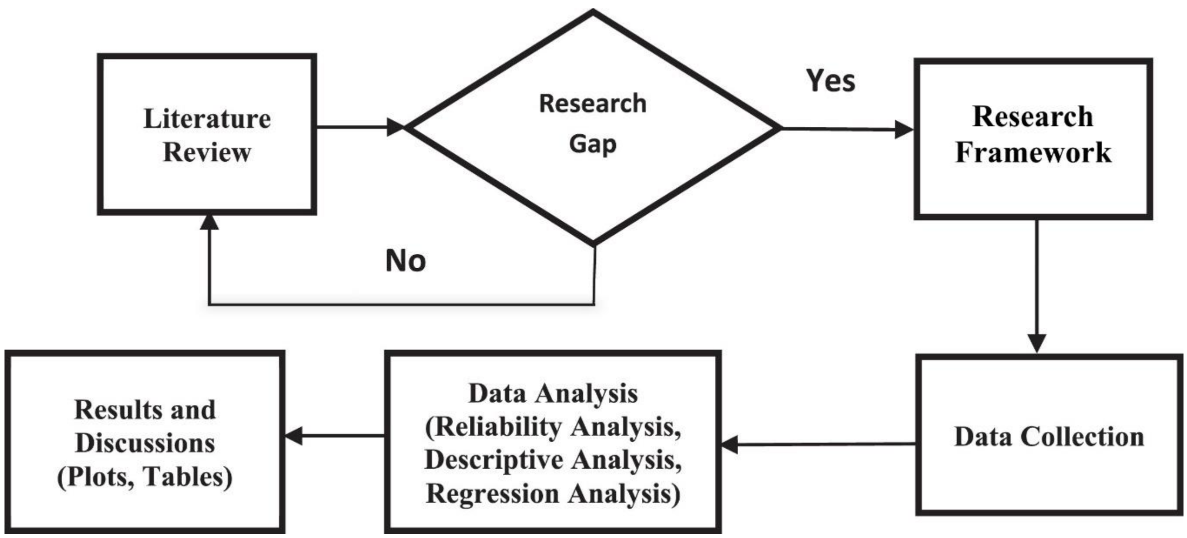


Figure 1: Research Process (Islam, Rayhan & Majumder 2022)

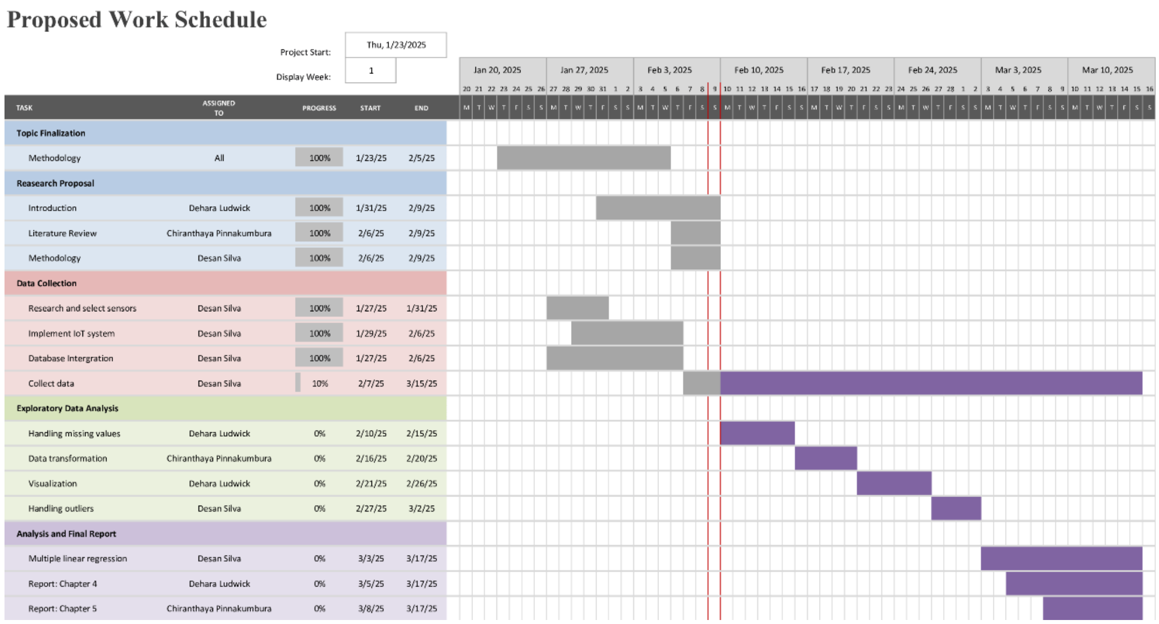


Figure 2: Gantt chart on work schedule

Chapter 2: Literature Review

2.1 Introduction to the research theme

The theme of this research explores the relationship between temperature and environmental factors such as humidity, UV intensity and light intensity localized to a single point of a university dining area. The focus of this research is to analyze the associations between these factors to create a predictive analysis that can be used to gain valuable insights.

2.2 Theoretical explanation about the Key Words in the Topic

2.2.1 Internet of Things (IoT)

The Internet of Things (IoT) is referred to as a network of physical devices, appliances, and other physical objects that are embedded with sensors, software, and network connectivity, which facilitates collection and sharing data among each other (IBM 2023). In this research, a collection of sensors and a microcontroller is used to gather and transmit data.

2.2.2 Predictive analysis

Predictive analysis is defined as the use of data to predict future trends and events. It is the use of statistical techniques and models to forecast outcomes based on existing data (Halton 2024; Tableau 2025).

2.2.3 Ambient environmental factors (Temperature, Humidity, UV level, Light level)

Environmental factors refer to the variables that influence the environment and climate (Werndl 2016). Temperature, humidity, UV intensity and luminosity in the context of an open commercial building area, determines thermal comfort and regulation (ASHRAE 2023) and can be analyzed to identify associations between them.

2.2.4 Early year environmental conditions

The environmental parameters measured at the beginning of a year constitute a small time frame where climate trends can be neglected due to its miniscule effect (Werndl 2016). The research focuses on this specific time frame to isolate these seasonal variations and neglect them when obtaining measurements.

2.3 Findings by other researchers

IoT based systems in the environmental context have enhanced the research activity in this field due to the low cost data collection methods (Polymeni et al. 2022). Various fields such as pollution and weather reporting, water and air quality assessments, as well as temperature monitoring systems have been implemented (Polymeni et al. 2022).

Researches conducted by Hidayat & Soekirno (2021) show that due to rapid urbanization, the urban heat island (UHI) effect has a detrimental impact on residential and commercial environments. The IoT setup consists of temperature sensors, a microcontroller, memory card and a Wi-Fi communication modem. UHI intensity information is used to predict and anticipate the impact of the UHI index, and the results obtained by this research can be used to benefit human health aspects through thermal information.

Temperature data collected by IoT data is mainly utilized for predictive control of HVAC systems in indoor commercial environments (Carli et al. 2020). This research focuses on automatically controlling HVAC systems based on the variables affecting the predicted mean vote (PMV), which is a measure of thermal comfort (ASHRAE 2023). Smart HVAC systems based on IoT integration also use time-series forecasting and reinforcement learning to increase accuracy (Zhuang et al. 2023).

The above researches conducted ensure valuable information used for enhancing thermal comfort, as well as overall human health through IoT based temperature analysis setups. Therefore, this study focuses on the above mentioned factors in the setting of section 2.4.

2.4 The research gap

While IoT based environmental monitoring is used widely in commercial buildings and industrial applications, The analysis of thermal variations among dining spaces is underexplored, due to studies mainly focusing on HVAC control instead of open spaces. The sample is further localized into a central point in the NIC dining area which is a unique area. Due to high foot traffic and fluctuating occupancy, the variations in ambient environmental conditions could provide relevant information on thermal associations.

2.5 Table for Variables, their definitions and sources

Variable	Definition	Source
Temperature	Ambient air temperature measured in degrees Celsius (°C)	HTU21D Sensor (Adafruit 2013)
Humidity	Relative humidity percentage in the environment	
UV intensity	Level of ultraviolet radiation in the environment	GUVA-S12SD Sensor (Roithner LaserTechnik 2011)
Light Intensity/ Luminosity	Intensity of light in the environment, measured in lux	BH1750FVI Sensor (ROHM 2011)

Table 1: Variables, definitions and sources

Chapter 3: Methodology

3.1 Introduction

This chapter outlines the research methodology adopted to analyze data and obtain solutions to the research questions. Sampling, data collection and analysis methods will be thoroughly discussed.

3.2 Population, sample and Sampling technique

3.2.1 Population

The population consists of independent readings obtained from the centroid of the canteen area, as shown in fig.3 . The schematics of the first floor of the building and the center point were calculated through AutoCAD. The time interval that the population data is collected is regular readings in the range 9:00 a.m. to 4:00 p.m. from Monday to Friday. in the time period of 07/02/2025 to 17/03/2025. The location and time frame were established to avoid complication of the model due to clustering from location variations, as well as to avoid effects from climate changes. Although a time frame is assigned, the data is assumed to be independent due to the comparatively short period of time.

3.2.2 Sampling Technique

According to Cochran (1977), The most effective sampling method to obtain independent and non-biased data is simple random sampling (SRS) which is required to perform an accurate linear regression model.

Each sample, X_i is selected independently from the population, with a random sampling interval. When the total number of possible data points (population) is given by N, the probability of being selected is given as:

$$\mathrm{P}(\mathrm{X_i}\ \text{is selected}) = \frac{1}{\mathrm{N}}$$

Where,

$\mathrm{N} = \text{population}\ (\mathit{Total\ data\ points\ possible})$

$X_i = \text{measured sensor readings}(\mathit{Temperature}, \mathit{Humidity}, \mathit{UV}, \mathit{Light\ intensity})$

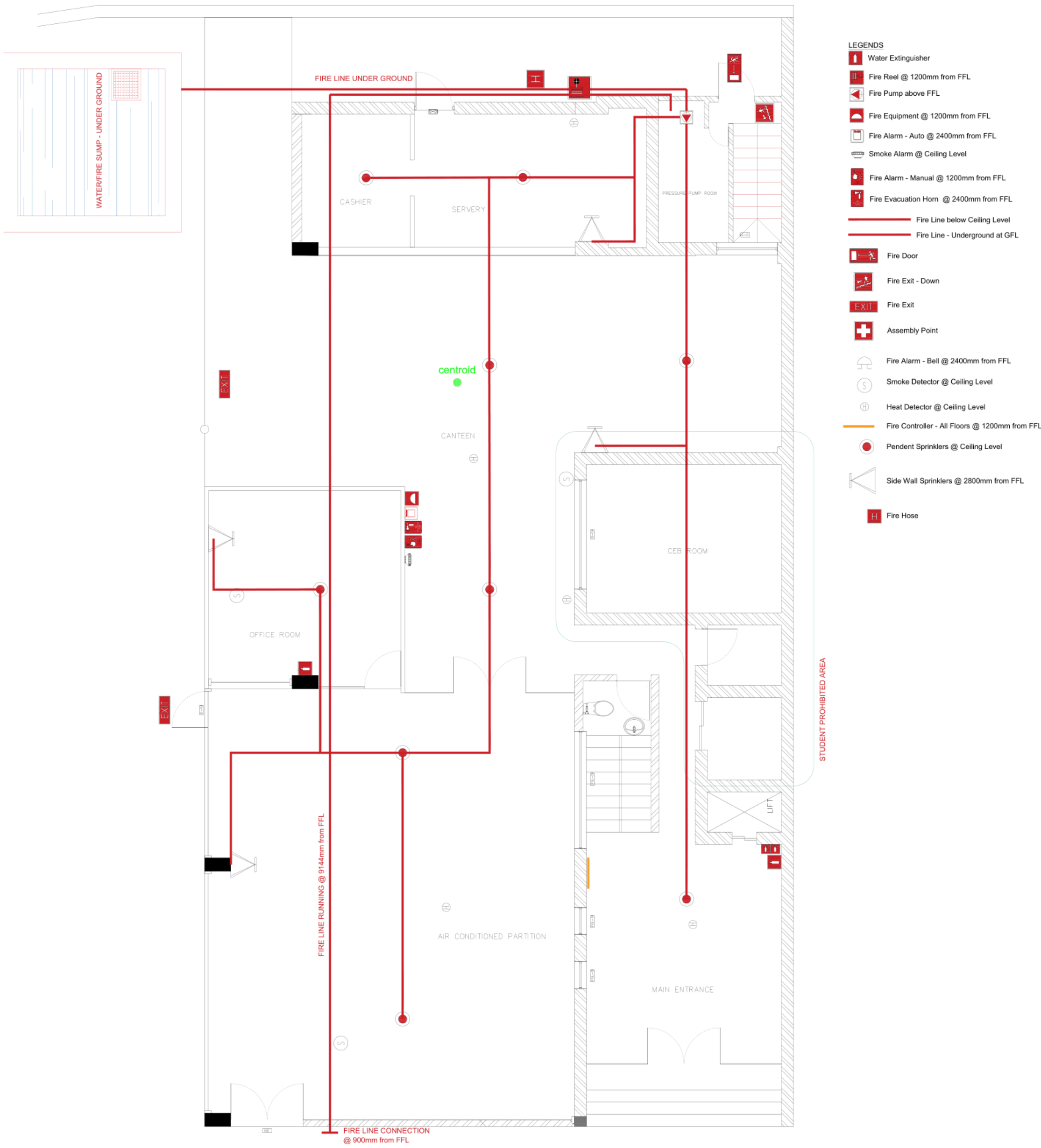
3.2.3 Sample

To ensure independence, a large amount of data has to be taken from the population while being reasonable. According to (Cochran 1977; Bhandari 2020), The number of samples of an unbiased random sample is inversely proportional to variance, which demonstrates that a large number of samples is required for accuracy.

For this research, a reasonable number of samples have to be obtained while adhering to the short sampling time frame. The total number of hours in the population time frame from 06/02/2025 – 16/03/2025 is given by 245 active hours or 882,000 seconds. A random sampling interval from a uniform distribution of 10 seconds and 5 minutes is assumed (U(10, 500) sec) which provides an expected rate of 259 seconds per sample. Dividing this amount from the total seconds gives the best case scenario of 3,400 readings. However, since the sampling period is random, the final amount of readings obtained for the research is 3,577 samples.

3.3 Type of Data to be collected and data sources

The data collected are Temperature (^{0}C), Humidity (%), UV intensity (mW/cm^{2}) and light intensity (lux), which are all collected from a single location from the canteen area as shown by the green point in fig.3. The point is taken using the MASSPROP command in AutoCAD from a schematic of the ground floor of the building and the readings are displayed in fig.4 below. It is assumed that variations in reading throughout the entire area are negligible. This avoids clustering and avoids complicated sampling methods or analysis methods (Cochran 1977).



Ground Floor
Existing

Figure 3: NIC ground floor plan with center point (AutoCAD 2023)

```
 ----------------   REGIONS   ----------------
Area:                     112083355.1660
Perimeter:                62191.6335
Bounding box:          X: 57009.8524  --  71329.8524
                       Y: 68889.7047  --  85647.2047
Centroid:              X: 63280.4306
                       Y: 76590.9252
Moments of inertia:    X: 6.5884E+17
                       Y: 4.5055E+17
Product of inertia:   XY: -5.4281E+17
Radii of gyration:     X: 76669.0548
                       Y: 63401.7386
Principal moments and X-Y directions about centroid:
                       I: 1069031722251488 along [0.8398 -0.5429]
                       J: 1995517956088544 along [0.5429 0.8398]
```

Figure 4: Readings obtained from AutoCAD MASSPROP command (AutoCAD 2023)

As mentioned in section 3.2 above, a reading is obtained from a random interval in the uniform range U(10, 500) while also taking the time limits set by the population into consideration.

3.4 Data collection tools and plan

The data is collected through an IoT setup which consists of sensors and a microcontroller unit (MCU) to read and transmit data in Json format to an AWS broker, which facilitates the transport of data to a local device controlled by the user through an MQTT connection, then sends the data into a MongoDB database. The specifications of the sensors and microcontroller are shown in table 2 below[1] (Espressif Systems 2023; Adafruit 2013; Roithner LaserTechnik 2011; ROHM 2011).

Device name	Type	Purpose	Specifications
ESP32-DevKitC V2	MCU	Run python script to collect and transmit data	38 GPIO pins, 3v3 logic level, 2.4GHz Wi-Fi/BLE
HTU21D	Sensor	Obtain temperature and humidity readings	Temperature resolution: 0.01^0C Temperature accuracy: $\pm 0.3^0C$ Humidity resolution: $0.04\%\ RH$ Humidity accuracy: $\pm 2\%\ RH$
BH1750FV1	Sensor	Obtain light intensity readings	Resolution: $1\ lux$ Accuracy: $\pm 20\%$
GUVA-S12SD	Sensor	Obtain UV intensity readings	Sensitivity: $0.14\ A/W$ Accuracy: $\pm 15\%$ Active area: $0.076 mm^2$

Table 2: Data Collection tools and specifications

[1] All datasheets can be found in the attachments section

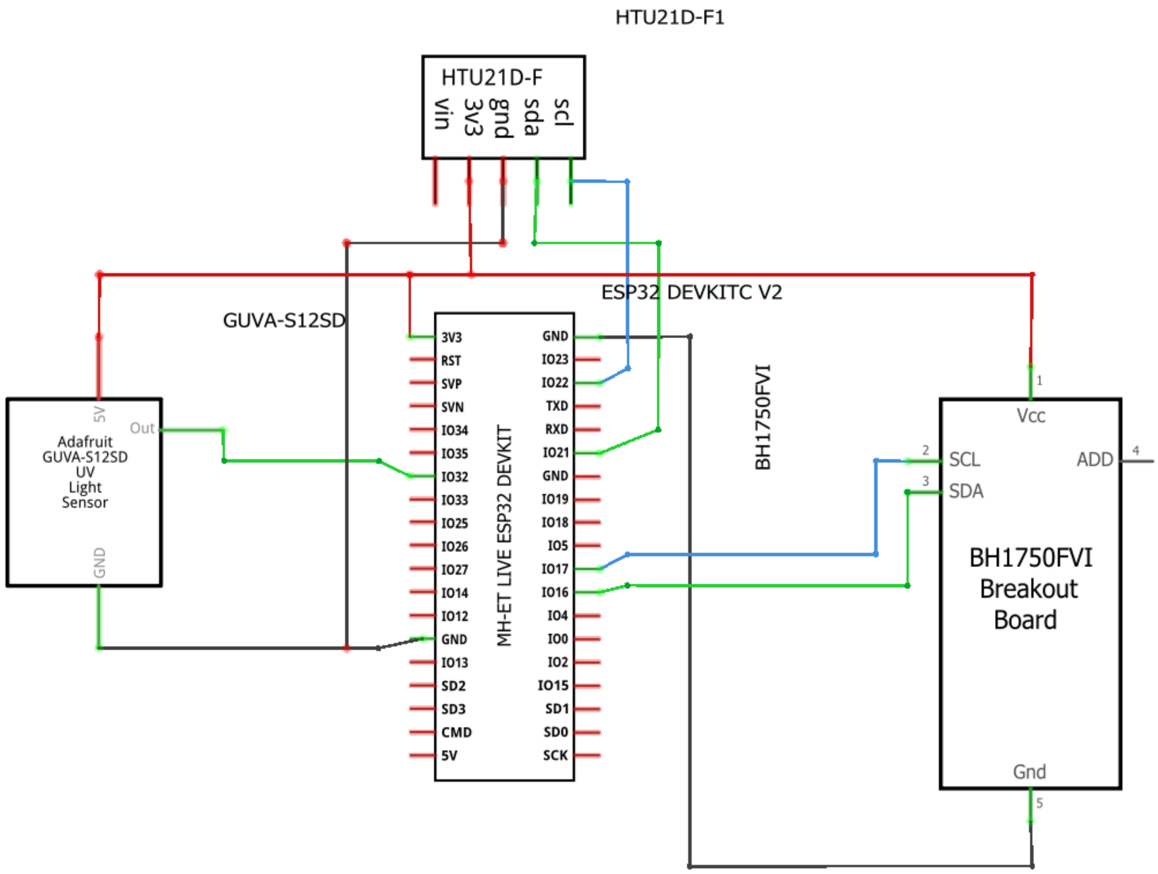


Figure 5: IoT setup schematic (Fritzing 0.9.3b.64)

The schematic of the entire setup is shown in fig.5 above (created through the fritzing application). The micropython framework is used along with Thonny IDE for programming the relevant scripts. And the connection handling and data collection/transmission is done through external python scripts[2].

[2] https://github.com/DesanSilva/Temperature-Analysis

The setup is powered through a 5V USB power bank supply which ensures safe power due to the ESP32 LDO regulator (Pieters 2023). It is positioned at the center point determined in section 3.3 and continuously run throughout the set time frame.

It collects data from a random sampling interval of 10-300 seconds, ensuring non periodic data collection to obtain independent samples.

3.5 Conceptual Framework

The key concepts considered in this research are data collection, data storage, data analysis and interpretation. The following data flow diagram in fig. 6 gives an overview of the data pipeline implemented.

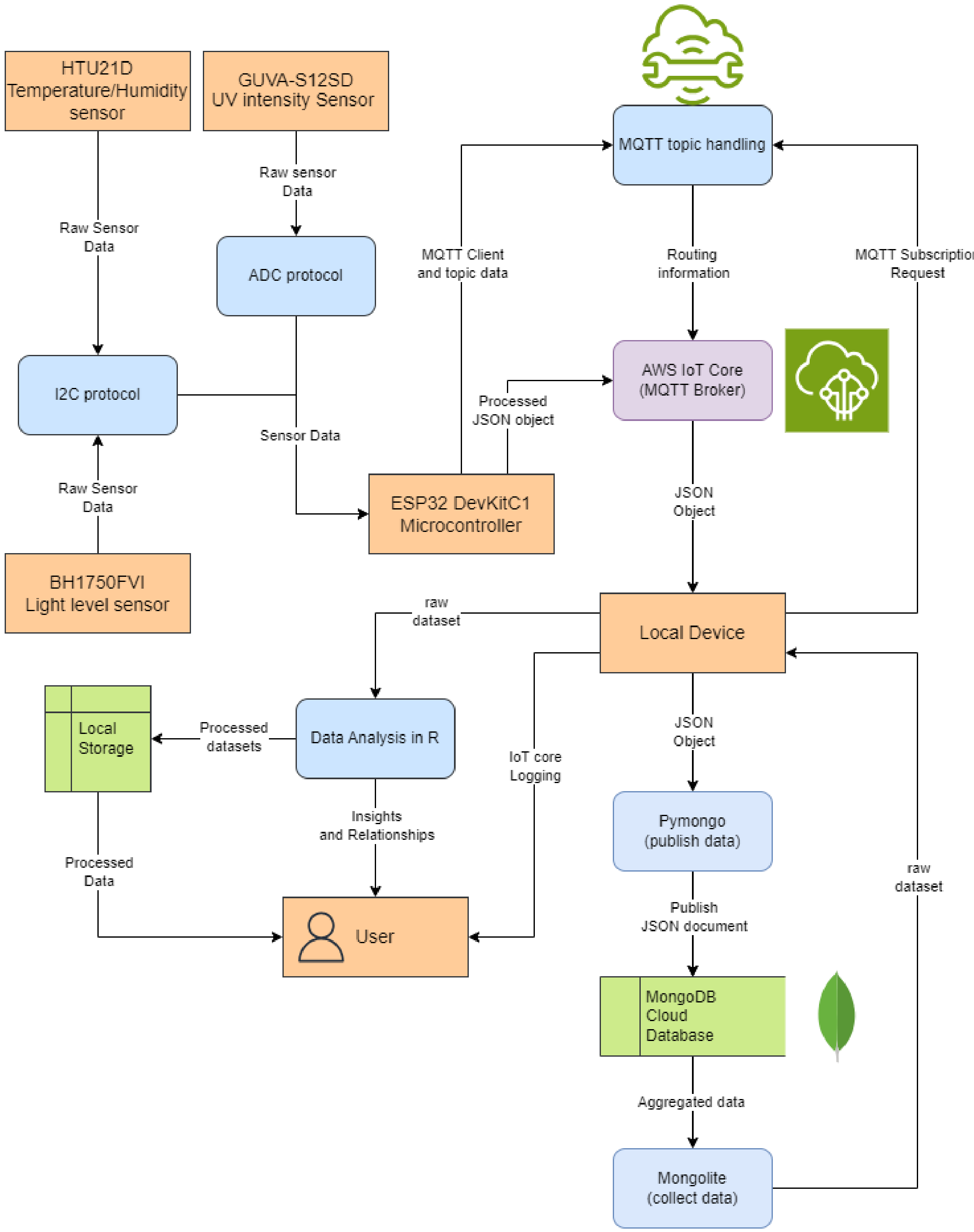


Figure 6: Dataflow diagram (Hidayat & Soekirno 2021)

3.6 Hypothesis

Assuming a Linear relationship for the population given by the following equation, the null and alternative hypotheses can be derived (Cochran 1977). Which then can be tested for the sample data.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

Where,

$Y \ = Dependent\ variable - Temperature$

$\beta_0 = y\ intercept\ term$

$X_1, X_2, X_3 = independent\ variables\ (Humidity,\ Light\ intensity,\ UV\ intensity)$

$\beta_1, \beta_2, \beta_3 = \ regression\ coefficients\ of\ Humidity,\ Light\ intensity,\ UV\ intensity$

$\epsilon = error\ term/residuals$

Null Hypothesis (H_0)

None of the independent variables: humidity, UV intensity and light intensity have a statistically significant relationship with temperature.

$$H_0 \colon \beta_0 = \beta_1 = \beta_2 = \beta_3 = 0$$

Alternate Hypothesis (H_1)

At least one of the independent variables has a statistically significant relationship with temperature.

$$H_1 \colon \exists\ \beta_i \neq 0, for\ at\ least\ one\ i \in \{1, 2, 3\}$$

Hypotheses for individual predictors

To identify key factors if the null hypothesis is not rejected, individual predictors are tested to test for a significant relationship between each variable with temperature.

$for\ variable\ X_1\ (Humidity)$:

$$H_0: \beta_1 = 0, \qquad H_1: \beta_1 \neq 0$$

$for\ variable\ X_2\ (Light\ intensity)$:

$$H_0: \beta_2 = 0, \qquad H_1: \beta_2 \neq 0$$

$for\ variable\ X_3\ (UV\ intensity)$:

$$H_0: \beta_3 = 0, \qquad H_1: \beta_3 \neq 0$$

3.7 Operationalization Table

Variable	Definition	Indicators	Measures	References
Temperature	The degree of heat present in the environment, measured in Celsius	Ambient temperature detected by the HTU21D sensor	Unit: ^{0}C Range: -40^{0}C to 125^{0}C Resolution: 0.01^{0} C Accuracy: ±0.3^{0} C	HTU21D datasheet (Adafruit 2013; Paul et al. 2018)
Humidity	The amount of water vapor present in the air, expressed as a percentage of relative humidity	Air moisture detected by the HTU21D sensor	Unit: %RH Range: 0–100% RH Resolution:0.04%RH Accuracy: ±2% RH	
Ultraviolet (UV) intensity	The strength of ultraviolet (UV) radiation in the environment	UV light detected by the GUVA-S12SD sensor	Unit: lux (lx) Range: 1–65535 lux Resolution: 1 lux Accuracy: ±20%	GUVA-S12SD Datasheet (Roithner LaserTechnik 2011; Paul et al. 2018)
Light intensity	The intensity of visible light in the environment, measured in lux	Light intensity detected by the BH1750 sensor	Unit: mW/cm² Range: 240-370 nm Sensitivity:0.14A/W Accuracy: ±15%	BH1750FVI Datasheet (ROHM 2011; Paul et al. 2018)

Table 3: Operationalization Table

3.8 Methods of data analysis

The data stored in MongoDB is extracted to R through the mongolite package, which allows for preprocessing and further analysis.

An exploratory data analysis (EDA) is performed through R which assists in understanding the collected data.

Missing values are identified and handled accordingly. Duplicate values are identified and omitted. Univariate analysis is performed to understand each variable through summary statistics, and Multivariate analysis is done to identify the relationships between variables. This can be achieved through correlation testing and scatter plots. Boxplots are used to identify and handle outliers.

Data is transformed/scaled and normalized, due to the variables containing contradistinctive scales (Cochran 1977) as shown in the process in figure 8 below.

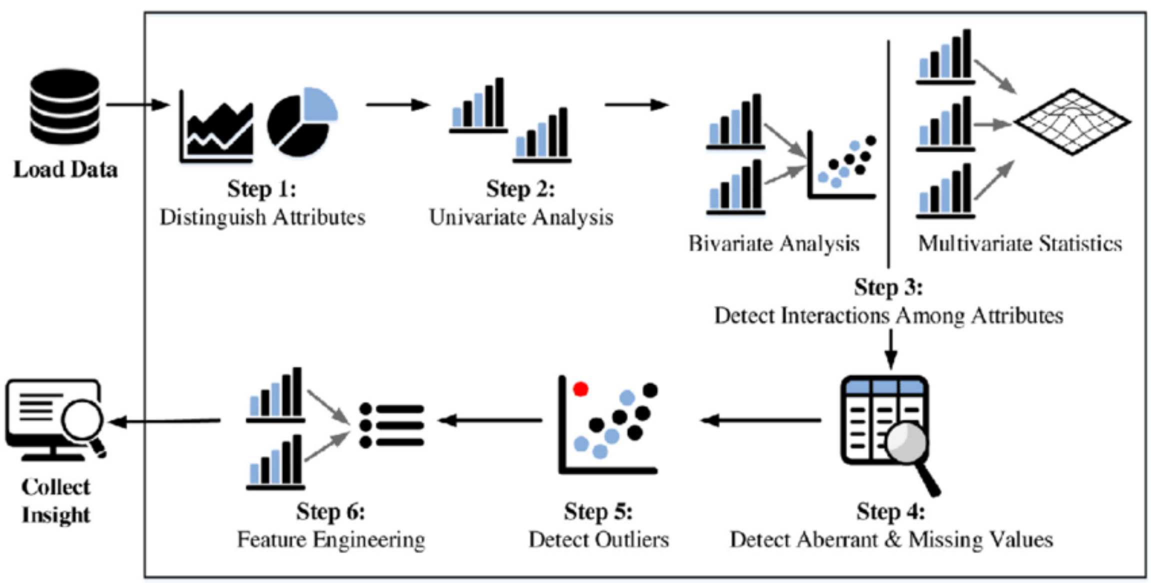


Figure 7: EDA workflow diagram (Ghosh et al. 2018)

The processed data is then used to perform a multiple linear regression , which can be used to verify or reject the hypothesis mentioned in section 3.6 (Mutombo & Numbi 2022; Huang, Miles & Zhang 2020; Cochran 1977). The following variables are utilized:

Dependent variable: Temperature

Independent variables: Humidity, UV intensity, Light intensity

After the regression is performed, assumptions are tested for linearity, independence, homoscedasticity, normality and no multicollinearity (Mutombo & Numbi 2022). Data is manipulated to fit these assumptions if possible. The sample data is split as 80% training data and 20% test data following Cochran's principles of random sampling (Cochran 1977). For the sample size used in this study, this percentage ensures that the training data is representative while keeping enough unseen data for evaluation. An 85%-25% could be used to improve the regression accuracy if a high variance among samples is observed (Cochran 1977). Finally, the model is fitted by measures such as MSE, RMSE, adjusted R^2 and correlation to check accuracy (Cochran 1977; Mutombo & Numbi 2022)

Chapter 4: Data Analysis

4.1 Data Analysis

The data obtained from MongoDB through the mongolite package is saved as a data frame. After saving, the file can be viewed as shown in fig.7 below.

	temperature <dbl>	humidity <dbl>	light_level <dbl>	uv_level <dbl>
1	26.57402	69.48904	95.00000	6.222222
2	26.90650	68.37515	95.83333	6.222222
3	27.17463	67.51303	92.50000	6.256411
4	27.38913	66.36099	96.66666	6.222222
5	27.55000	65.36154	95.83333	6.188035
6	27.63580	64.55283	92.50000	6.188035

Figure 8: Raw Data from MongoDB

An immediate observation can be made that all variables are quantitative. Light intensity and UV intensity are given by the names light_level and uv_level, which can be used as it is for convenience. Before performing an EDA, the raw data is split into a training dataset containing 80% of samples and a testing dataset containing 20% samples.

4.1.2 Exploratory Data Analysis (EDA)

4.1.2.1 Descriptive Statistics

After loading the training dataset into R, an overview of the data and summary statistics are obtained.

```
Rows: 2,862
Columns: 4
$ temperature <dbl> 28.09480, 29.63068, 27.45348, 28.68892, 28.44019, 29.81504, 29.34109, 28.27272, 29.26602, 28.62238, 28.75046, 29.3…
$ humidity    <dbl> 72.61815, 69.20676, 74.40237, 73.52837, 69.96970, 66.73166, 68.58878, 70.01513, 69.35934, 71.61094, 71.96827, 67.5…
$ light_level <dbl> 200.39393, 481.66670, 128.69916, 330.39322, 312.50000, 511.80674, 452.50000, 455.89219, 460.00000, 430.34168, 257.…
$ uv_level    <dbl> 6.104336, 6.101721, 5.970501, 5.903404, 6.134526, 6.101939, 6.101721, 6.146071, 6.134526, 5.935547, 6.111038, 6.10…
```

Figure 9: dplyr::glimpse() output

```
  temperature       humidity       light_level         uv_level
 Min.   :25.76   Min.   :57.25   Min.   :  7.148   Min.   :5.052
 1st Qu.:28.41   1st Qu.:67.56   1st Qu.:159.342   1st Qu.:6.015
 Median :29.02   Median :69.63   Median :291.250   Median :6.091
 Mean   :28.86   Mean   :70.18   Mean   :325.680   Mean   :6.081
 3rd Qu.:29.39   3rd Qu.:72.27   3rd Qu.:458.410   3rd Qu.:6.135
 Max.   :31.28   Max.   :83.71   Max.   :975.333   Max.   :6.922
```

Figure 10: summary output

These show data types, and descriptive statistics such as mean, median, quartiles, minimum and maximum values. At this stage it is still not clear to identify if the data is suitable for training a statistical model.

4.1.2.2 Univariate Analysis

Histograms, and density plots are used to visualize the distributions of each variable and the ranges that they lie in. Q-Q plots are used to identify whether variables follow a normal distribution, and boxplots are used to identify outliers.

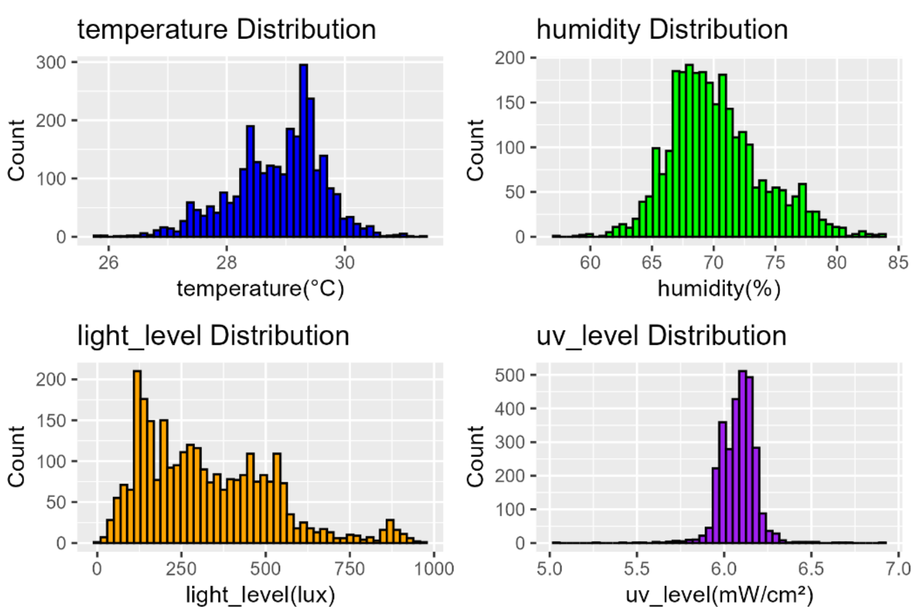


Figure 11: Histograms of raw data

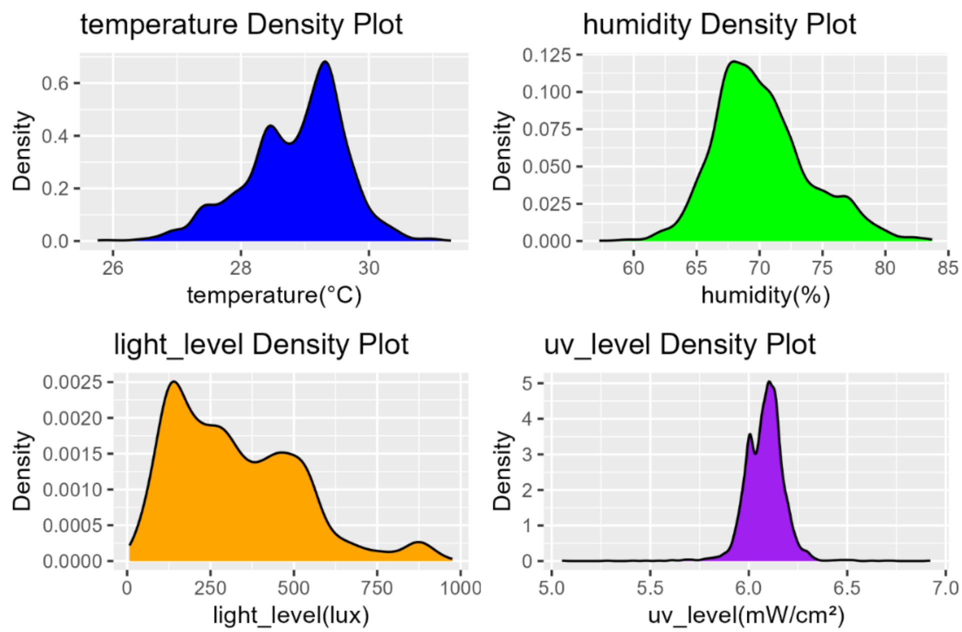


Figure 12: Density plots of raw data

The preferred type of data in this scenario is a normal distribution (bell shaped curve centered around the mean) with single peaks (unimodal) since it facilitates parametric models.

The data here shown by figure 11 and 23 show multiple peaks in some locations, as well as uneven values which indicate outliers. Outliers can be further confirmed by boxplots and other methods in the data wrangling process.

The Central Limit Theorem (CLT) states that if you take sufficiently large samples from a population, the samples' means will be normally distributed, even if the population isn't normally distributed, and real world continuous variables tend to be normally distributed due to this phenomenon (Turney 2022).

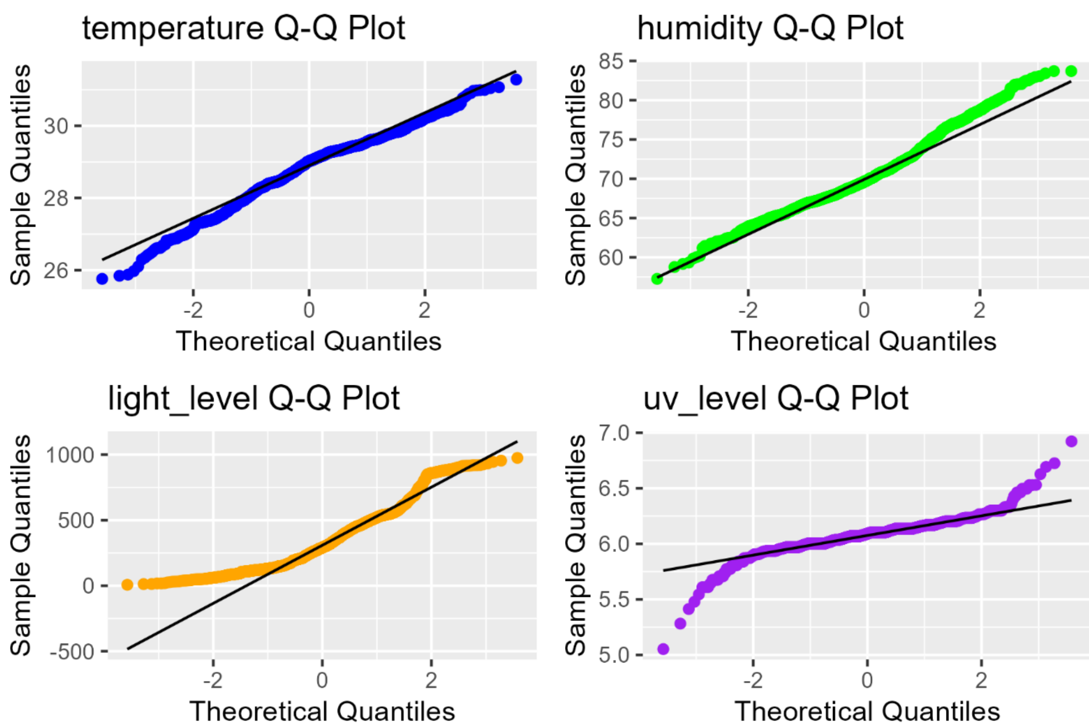


Figure 13: Q-Q Plots of raw data

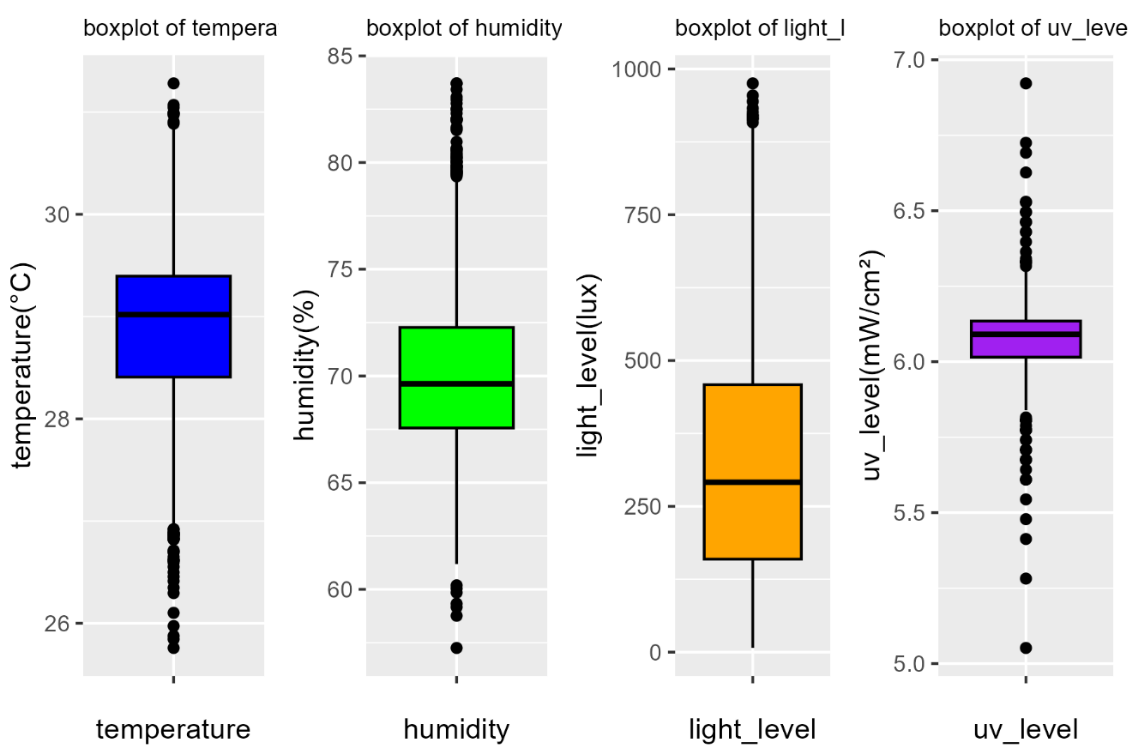


Figure 14: Boxplots of raw data

Figure 13 depicts that variables do not follow normal distributions, with light_level and uv_level having the most deviated plots.

Figure 14 depicts that all variables contain outliers which have to be handled accordingly.

4.1.2.3 Bivariate Analysis

This step gives correlation coefficients of each variable as well as scatter plots of all variables with each other.

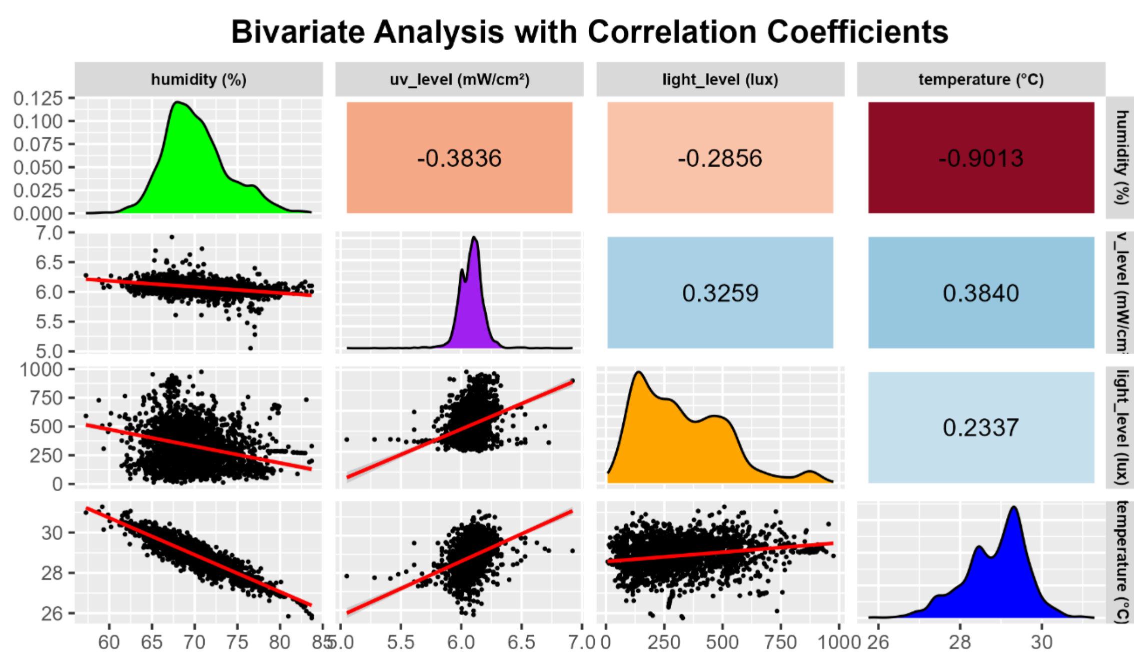


Figure 15: Bivariate analysis and correlation coefficients of raw data

The significant plots in this image are the scatter plots on the final row and the coefficients on the final column. Scatter plots show that variables roughly have a linear relationship with humidity showing the most linearity and light_level the least.

Correlation coefficients lie in the range -1 to 1, where 0 shows no correlation and -1 and 1 show perfect negative and positive correlation respectively (Nickolas 2024). The values in figure 15 depict that temperature and humidity have an extremely high negative correlation while uv_level has moderate positive correlation and light_level has very low positive correlation with temperature.

4.1.2.4 Multivariate Analysis

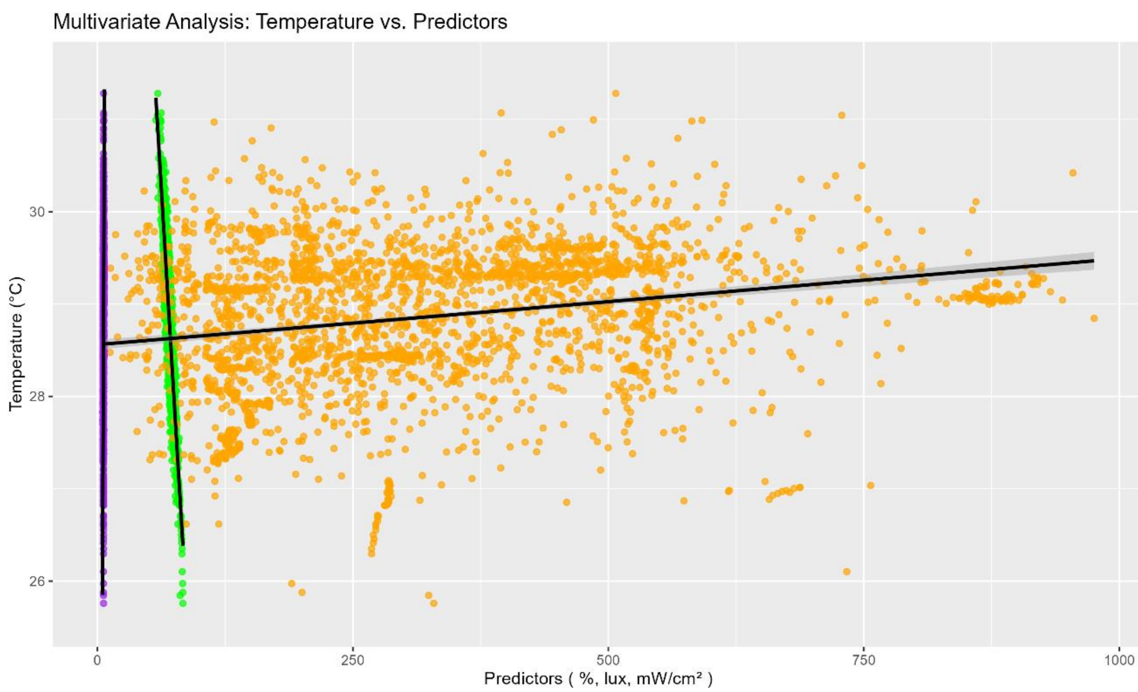


Figure 16: Multivariate Plot of raw data

The multivariate analysis plot in figure 16 shows that the predictor variables should be scaled before fitting a statistical model since the variables have varied ranges due to different units.

4.1.2.5 Handling Null values and Outliers

Missing or duplicate data and outliers can lead to skewed data and incorrect conclusions (Djizmedjian 2025). Therefore, they must be handled accordingly.

No null values or duplicate rows were identified during the analysis, but outliers exist for every variable, as implied by the boxplots in 4.1.2.2 univariate analysis above.

The Interquartile range method (IQR method) is used to identify outliers. The bounds taken for outlier identification are as follows:

upper bound = $Q3 + (1.5 * IQR)$

lower bound = $Q1 - (1.5 * IQR)$

where; Q1 = first quartile, Q3 = third quartile, IQR = Q3 – Q1

```
Outliers in temperature :

 25.75891 25.84471 25.87689 25.97341 26.10212 26.29517 26.34879 26.41315 26.45605 26.49895 26.55257 26.59547 26.61692 26.61692 26.61692 26.64909
26.69199 26.70272 26.71345 26.8207 26.8207 26.8207 26.83142 26.83142 26.85287 26.85287 26.8636 26.86889 26.87432 26.8757 26.88505 26.88505
26.88505 26.91722 26.92095 30.88688 30.90696 30.97131 30.98203 30.9911 30.99421 31.04379 31.07013 31.28083

Total: 44

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

Outliers in humidity :

 57.25498 58.7625 59.16603 59.32746 59.84796 60.03791 60.18885 79.35385 79.42252 79.46829 79.48355 79.52933 79.54511 79.5751 79.59799 79.60562
79.68954 79.72006 79.78873 79.81161 79.81924 79.82687 79.89554 80.00998 80.02524 80.0405 80.10097 80.20834 80.2236 80.26938 80.33804 80.39908
80.47537 80.57455 80.59744 80.65848 80.67374 80.97128 81.5206 81.58163 81.65793 81.9631 81.98599 82.0394 82.0394 82.05466 82.30643 82.4819
82.52768 82.77182 82.9473 83.077 83.42795 83.71024 83.71024

Total: 55

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

Outliers in light_level :

 908.3333 914.1666 914.1666 914.9999 915.8333 915.8333 916.6666 919.9999 919.9999 919.9999 920.8333 925.8333 932.5824 944.3775 954.6124 975.3328

Total: 16

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

Outliers in uv_level :

 5.051963 5.281598 5.412817 5.478427 5.544037 5.609647 5.609647 5.609647 5.642451 5.675257 5.675257 5.675257 5.675257 5.708062 5.708062 5.708062
5.708062 5.740867 5.740867 5.773672 5.773672 5.773672 5.773672 5.773672 5.787886 5.806477 5.806477 5.806477 5.806477 5.806477 5.806477 5.806477
5.815239 6.316936 6.326874 6.331357 6.331357 6.331357 6.331357 6.331357 6.341057 6.364161 6.396966 6.42977 6.42977 6.462575 6.462575 6.462575
6.49538 6.495727 6.495727 6.528185 6.528185 6.529915 6.6266 6.692211 6.725015 6.921845

Total: 58

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
```

Figure 17: Outlier analysis using IQR method

The values obtained for light_level are inconsistent with actual indoor light level readings. This could be due to accidental sunlight exposure or sensor malfunctions. Previously in the summary statistics (Figure 10) it was observed that the minimum value was less than 10. This is also impossible in a standard indoor environment and can be caused due to the sensor being obstructed. Since the lower end values do not appear but are still faulty, a range of 100-750 lux is approximated based on the light levels of other similar locations shown by Kumar (2023).

Transforming and Scaling data

The values obtained for temperature, humidity and uv_level cannot be removed or imputed since they are possible commonly occurring values. Therefore, the variables are transformed to better fit the model. Reciprocal transformation is selected as the choice of transformation since it is possible to make data more symmetric and manage extreme values, which are present here (Lee 2025). Using this transformation, a dataset $\{x_1, x_2, x_3, \dots, x_n\}$ of non-negative and non-zero values of x can be represented as $\{y_1, y_2, y_3, \dots, y_n\}$, where

$$y_i = \frac{1}{x_i}$$

According to (Rangi 2024), scaling is required after transforming since features are expected to have a uniform contribution to the model even though variables have varying scales . Scaling is also a method of normalization, but a separate normalization is required beforehand since outliers can drastically influence the scaled values. The dataset is scaled between the range [0, 1] which is a standard range used when performing min-max scaling (Rangi 2024). The equation below depicts how the data is transformed with this method.

$$y_i = \frac{x_i - \min(x)}{\max(x) - \min(x)}$$

The dataset after applying reciprocal transformation and min-max scaling in the range [0,1] is depicted by figure 18 below.

	temperature <dbl>	humidity <dbl>	light_level <dbl>	uv_level <dbl>
1	0.5290077	0.3305746	0.45144119	0.3070292
2	0.2597881	0.4535497	0.09158528	0.3084547
3	0.6503365	0.2707477	0.79472973	0.3815990
4	0.4214463	0.2996911	0.20897152	0.4202566
5	0.4659305	0.4250059	0.23037279	0.2906575
6	0.2293375	0.5506437	0.07648714	0.3083356

Figure 18: Dataset after normalization

Further Analysis

The visualizations done before preprocessing are done again to confirm that the data is properly scaled and normalized.

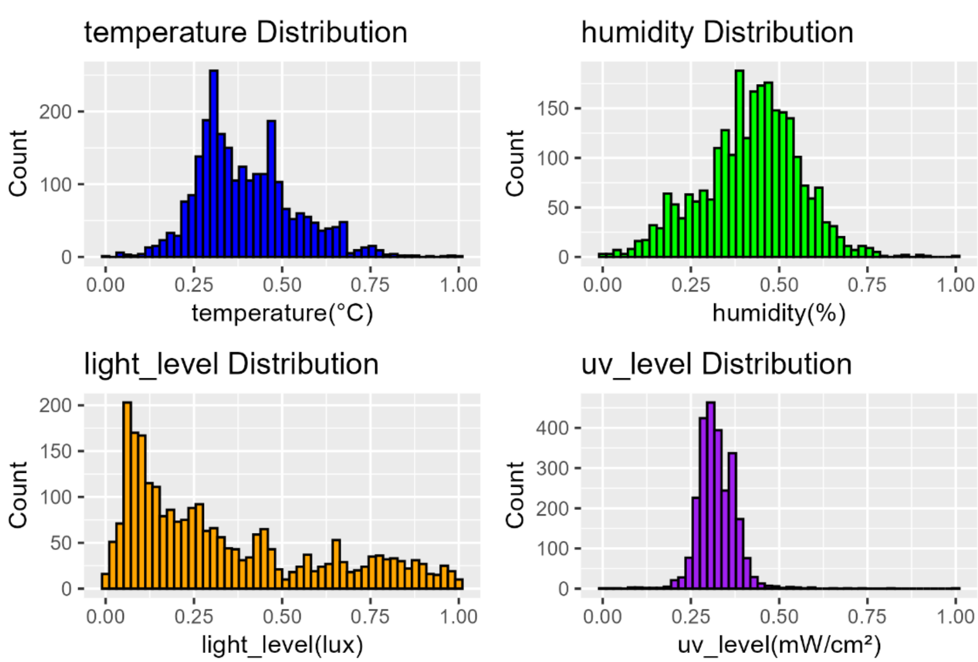


Figure 19: Histograms after preprocessing

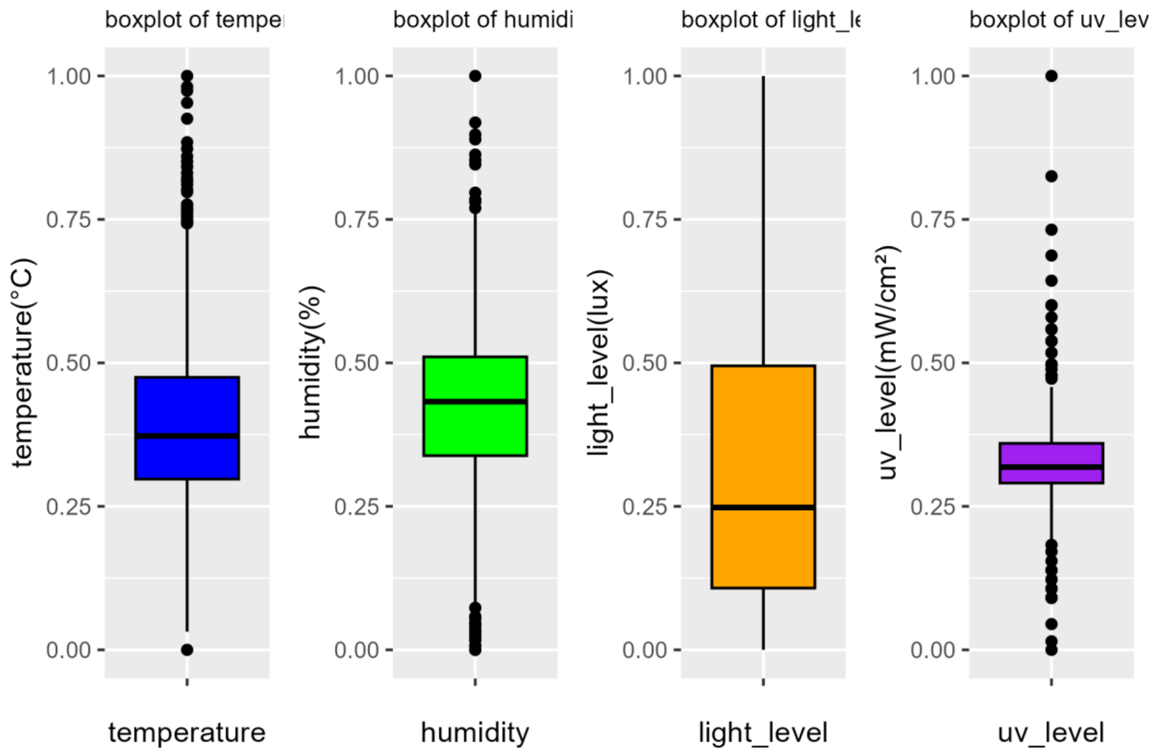


Figure 20: Boxplots after preprocessing

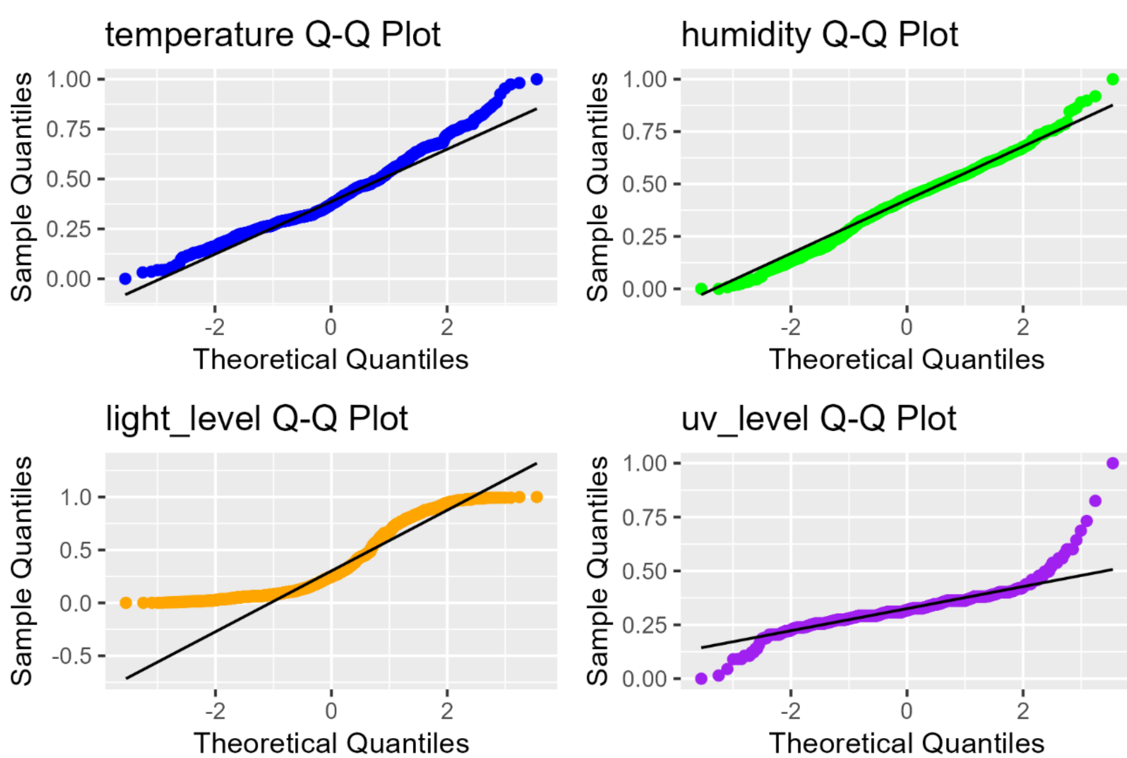


Figure 21: Q-Q plots after preprocessing

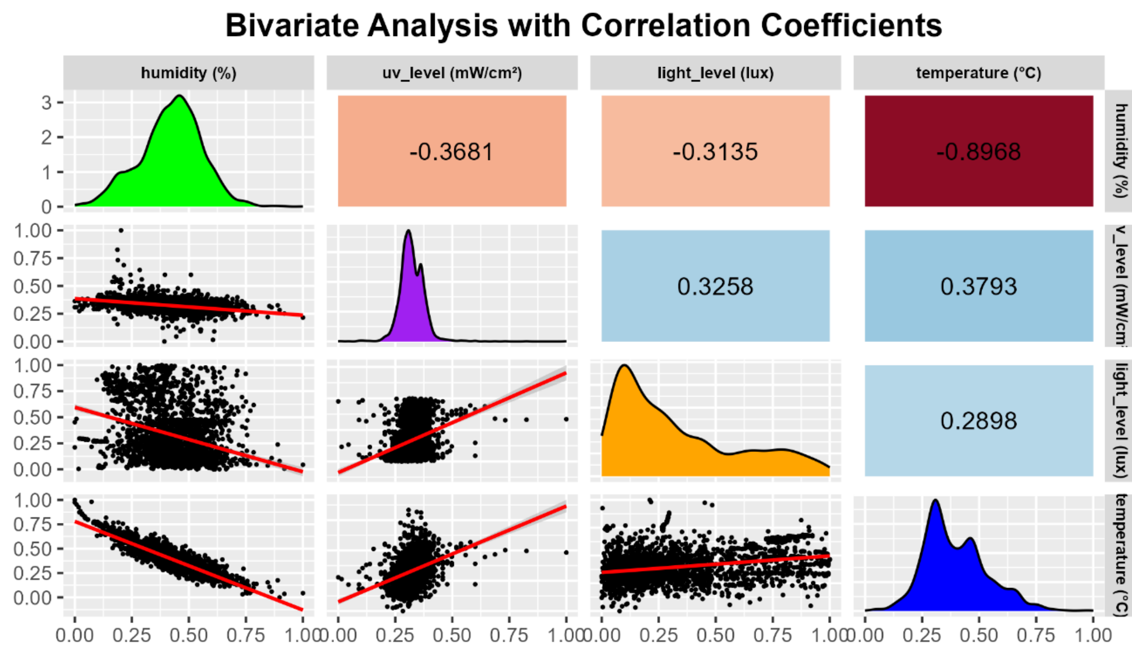


Figure 22: Bivariate analysis after preprocessing

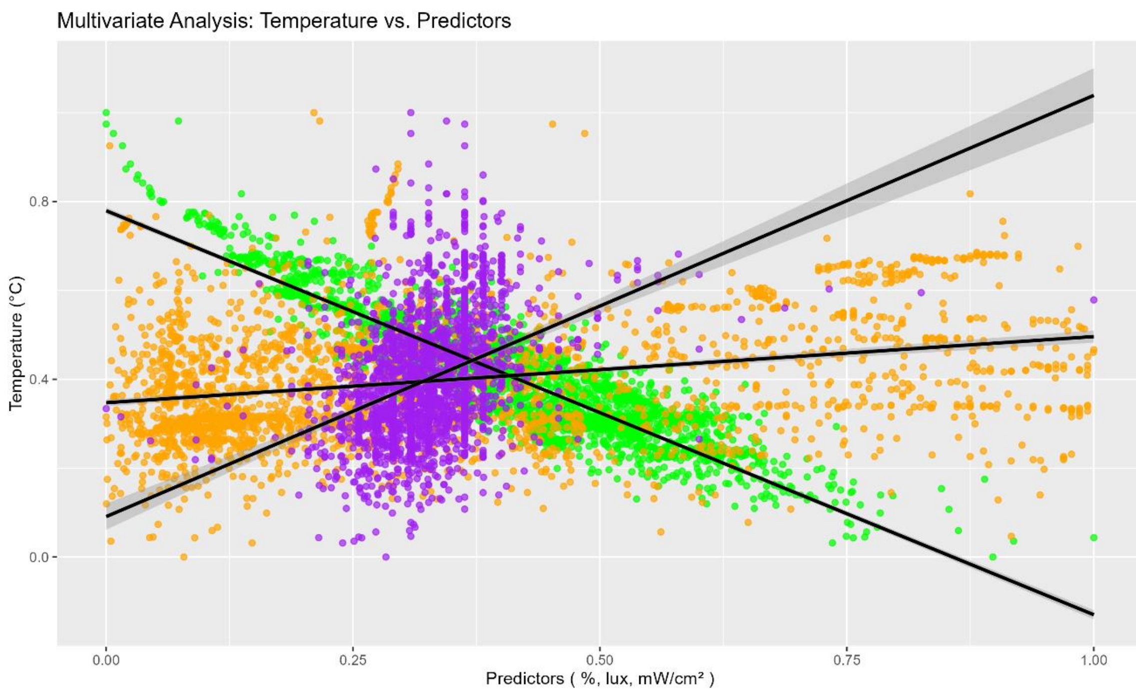


Figure 23: Multivariate analysis after preprocessing

The above data confirms that outliers have been removed from the light_level variable, and the dataset has been normalized and scaled successfully.

4.1.3 Multiple Linear Regression Model

Due to the associations identified in the EDA and as stated in the research methodology, a multiple linear regression model is determined to be the most convenient model for the research problem.

4.1.3.1 Model fitting and Interpretation

The regression model is represented using the equation mentioned in 3.6 above, and the coefficients as well as residual standard error are determined from the model summary.

```
Call:
lm(formula = temperature ~ humidity + light_level + uv_level,
    data = df)

Residuals:
      Min        1Q    Median        3Q       Max
-0.170796 -0.041737 -0.000319  0.043780  0.272123

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.724744   0.009842  73.636  < 2e-16 ***
humidity    -0.888851   0.009718 -91.469  < 2e-16 ***
light_level -0.002041   0.004844  -0.421    0.674
uv_level     0.144616   0.024062   6.010 2.12e-09 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.0611 on 2552 degrees of freedom
Multiple R-squared:  0.8071,    Adjusted R-squared:  0.8069
F-statistic:  3560 on 3 and 2552 DF,  p-value: < 2.2e-16
```

Figure 24: Regression Model summary

- The y intercept has a $p - value < 0.001$ (Signif.code: '***' [0, 0.001]). Therefore, the humidity variable shows a high statistically significant relationship with temperature at the 5% significance level. Therefore, a non-zero value exists for the intercept, $\beta_0 = 0.724737$.

- Humidity has a $\mathrm{p-value} < 0.001$ (Signif.code: '***' [0, 0.001]). Therefore, the humidity variable shows a high statistically significant relationship with temperature at the 5% significance level. The null hypothesis for X_1 predictor is rejected at 95% confidence level, $\beta_1 = -0.888851$

- light_level has a $\mathrm{p-value} > 0.05$ (Signif.code: ' ' [0.5, 1]). Therefore, the light_level variable shows no statistically significant relationship with temperature at the 5% significance level. The null hypothesis for X_2 predictor is accepted at 95% confidence level. The coefficient is approximately 0 but exact value is used for accuracy, $\beta_2 = -0.002041$

- uv_level has a $\mathrm{p-value} < 0.001$ (Signif.code: '***' [0, 0.001]). Therefore, the uv_level variable shows a high statistically significant relationship with temperature at the 5% significance level. The null hypothesis for X_2 predictor is rejected at 95% confidence level, $\beta_2 = 0.144616$

- Overall null hypothesis stated in 3.6 above is rejected. accept alternate hypothesis.

Conclusion

$$H_1: \exists\, \beta_i \neq 0, for\ at\ least\ one\ i \in \{1, 2, 3\}$$

Since the above statement is true, the final regression model is interpreted as:

$$\hat{Y} = (0.724744) + (-0.888851)X_1 + (-0.002041)X_2 + (0.144616)X_3 \pm (0.0611)$$

Where,

$\hat{Y} = Predicted\ scaled\ temperature$

$X_1, X_2, X_3 = respective\ predictor\ variables$

4.1.3.2 Linearity

The objective of this test is to determine if there is a linear relationship between predictors and the response (linearity)

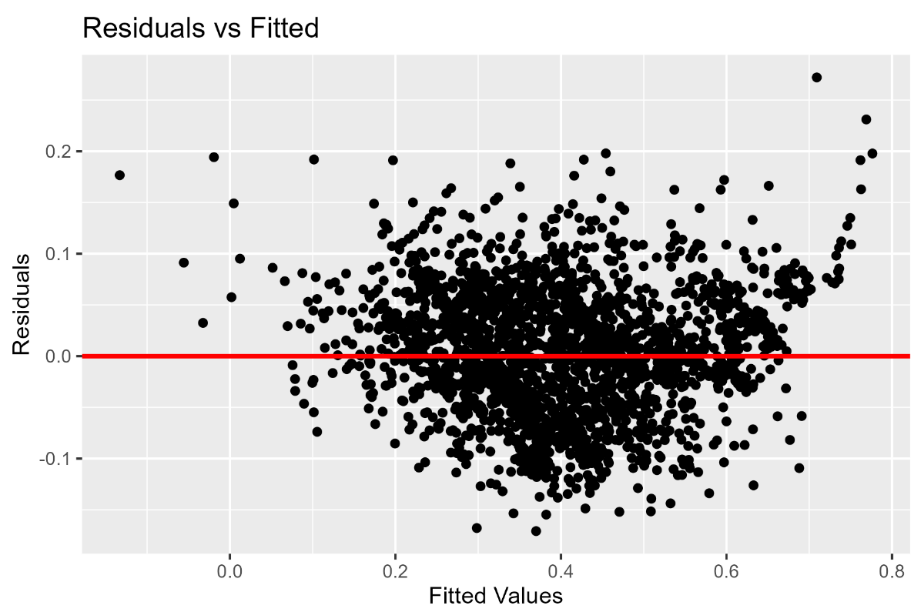


Figure 25: Residuals vs Fitted values

Residual values appear to be scattered randomly around the 0 value which confirms that there is a linear relationship between predictors and the response.

4.1.3.3 Independence

The Durbin-Watson test is used to measure independence between residuals of the model. The objective of this test is to determine if the error terms are independent or if they exhibit a pattern over time (autocorrelation) .

Hypotheses for Durbin-Watson test

H_0: residuals have no autocorrelation

H_1: true autocorrelation of residuals is greater than 0

```
        Durbin-Watson test

data:  model
DW = 1.9874, p-value = 0.3743
alternative hypothesis: true autocorrelation is greater than 0
```

Figure 26: Durbin-Watson independence test

Conclusion

H_0 is accepted at $\alpha = 0.05$ (95% confidence level) since $\text{p} - \text{value} > 0.05$.

Therefore, residuals have no autocorrelation.

4.1.3.4 Normality

The Shapiro-Wilk normality test is used to determine the normality of the model. The main objective is to determine whether the distribution of samples follows a normal distribution.

Hypotheses for Shapiro-Wilk test

H_0: residuals are normally distributed

H_1: residuals deviate from a normal distribution

```
        Shapiro-Wilk normality test

data:  model$residuals
W = 0.99488, p-value = 1.003e-07
```

Figure 27: Shapiro-Wilk normality test

Conclusion

H_0 is rejected at $\alpha = 0.05$ (95% confidence level) since $\mathrm{p} - \mathrm{value} < 0.05$.

Therefore residuals deviate from a normal distribution.

Q-Q plot of residuals

A Q-Q plot can be used to visualize the distribution of data with a theoretical normal distribution. It is used since the Shapiro-Wilk test can be sensitive to large datasets.

Figure 28: Q-Q plot of residuals

The residuals mostly follow a normal distribution visually. Given the Shapiro test's sensitivity to large sample sizes, the normality assumption can be reasonably satisfied using this plot.

4.1.3.5 Homoscedasticity

The Non-Constant Variance score test (ncvTest) is utilized to identify if the variance among residuals is constant in the model.

Hypotheses of the ncvTest

H_0: Homoscedasticity is present (residuals have constant variance)

H_1: Homoscedasticity is not present (residuals do not have constant variance)

```
Non-constant Variance Score Test
Variance formula: ~ fitted.values
Chisquare = 0.882819, Df = 1, p = 0.34743
```

Figure 29: ncvTest for homoscedasticity

Conclusion

H_0 is accepted at $\alpha = 0.05$ (95% confidence level) since $p - value < 0.05$.

Therefore, residuals have constant variance and the regression model is homoscedastic.

4.1.3.6 Multicollinearity

Multicollinearity is determined using Variance Inflation Factor (VIF) which measures the inflation of the variance of a regression coefficient due to other predictor variables.

```
humidity light_level    uv_level
1.215702    1.175799    1.226364
```

Figure 30: VIF values for multicollinearity

The ideal case is where $VIF = 1$ which has no multicollinearity. In this case $1 < VIF < 5$ which shows low multicollinearity but is acceptable in a regression model.

4.1.3.7 Model Evaluation and residual analysis

To use the regression model on the testing data, the data is first transformed in the same methods used for the train dataset (reciprocal transformation and min-max scaling). After it is transformed. The values are predicted and an approximation of the actual temperature is obtained by applying the same transformation inversely. Figure 31 shows the values after the predicted values have been obtained for the testing data.

	actual <dbl>	predicted <dbl>	approx_prediction <dbl>
1	0.3371420	0.4051469	27.99610
2	0.3136408	0.4365892	28.16972
3	0.4243635	0.4533900	28.26249
4	0.3961421	0.3944285	27.93691
5	0.3273257	0.3989909	27.96211
6	0.3866944	0.3310731	27.58707

Figure 31: Predicted data

Figure 32 shows the plot made between actual and predicted scaled data.

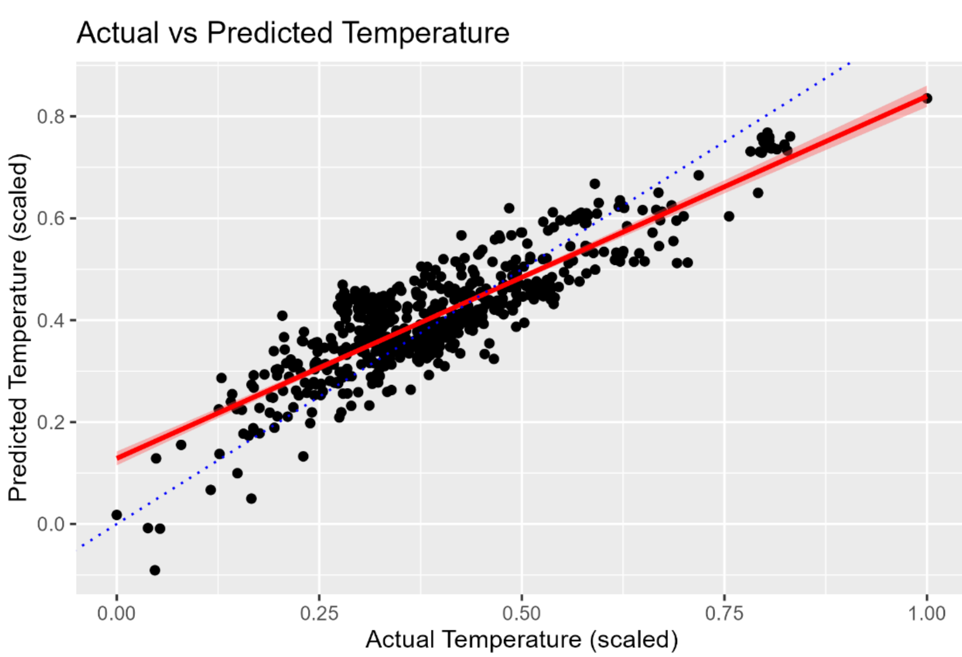


Figure 32: Actual vs predicted values

The blue dotted line shows the requirement for a perfect prediction, and the red line shows the actual performance of the model. There is no large deviation and it can be considered relatively accurate.

Residual Analysis

Residuals are calculated from the predicted values dataset by subtracting predicted values from the actual values, and they are plotted with the predicted temperature to interpret the data further.

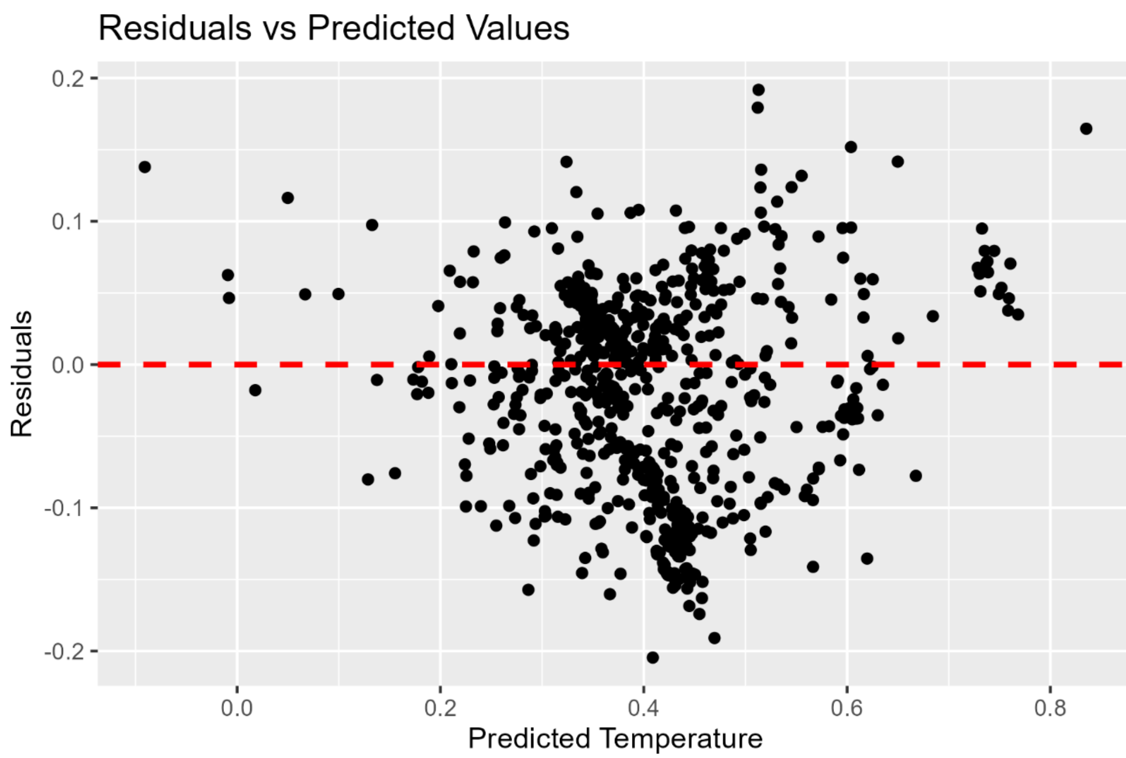


Figure 33: Residuals vs predicted values

The residuals of tested data are randomly scattered, which denotes a well fitted model and no clustering or systematic patterns.

4.1.3.8 Model Performance Metrics Interpretation

To test the accuracy of the model and how well it explains correlations between variables, performance metrics such as mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), R-squared and Adjusted R-squared are analyzed.

```
Model Performance Metrics:
--------------------------
- Mean Absolute Error (MAE):      0.058064
- Mean Squared Error (MSE):       0.005168
- Root Mean Squared Error (RMSE): 0.071890
- R-squared:                      0.807120
- Adjusted R-squared:             0.806893
```

Figure 34: Model Performance Metrics

Mean Absolute Error (MAE)

Mean absolute error is a measure of errors between paired observations expressing the same phenomenon (Wikipedia 2025). It shows how close the model’s predictions are to the actual values.

The value of 0.058064 obtained for this model suggests the average error in predictions is 0.05 units. In the scale of 0-1 which the model is in, this gives a 5.8% error which is not significant but also not perfect.

Mean Squared Error (MSE) and Root Mean Squared Error (RMSE)

Mean squared error is a measure of the average squared difference between the estimated values and the true value. And root mean squared error is the square root of this value which is easier to interpret (Wikipedia 2025). MSE = 0.005168 and RMSE = 0.071890 shows that predictions deviate by the approximately 0.07.

From the MAE value calculated earlier, it can be assumed that the deviation is in between the values given by both tests.

R-squared and Adjusted R-squared

R-squared measures the proportion of variance in the dependent variable explained by the independent variable while the Adjusted R-squared metric adjusts the R-squared value based on the number of predictors in the model. It accounts for the degrees of freedom, penalizing the addition of irrelevant predictors (GeeksforGeeks 2024).

This coefficient ranges from 0 to 1 where values closer to 1 indicate the variance dependent variable being explained more accurately by the independent variables.

The model has an R-squared value of 0.807120 which shows that the model explains more variance and is a better fit. The adjusted R-squared value of 0.806893 suggests that most predictor variables contribute meaningfully to the model (since the value is close to the R-squared value)

4.2 Findings and Interpretation

Through the EDA process, it is seen that the data obtained is adequate to build a multiple linear regression model.

The MLR model gives a relationship of

$$\hat{Y} = (0.724744) + (-0.888851)X_1 + (-0.002041)X_2 + (0.144616)X_3 \pm (0.0611)$$

Assumptions of linearity, homoscedasticity and multicollinearity are satisfied, while normality can be assumed through visual interpretations.

Therefore, the model is valid and the performance metrics of MAE, MSE, RMSE, R^2 and Adjusted R^2 confirm that the model has high accuracy and fits well.

Chapter 5: Discussion and Recommendations

5.1 Discussion

The findings of the study highlight the significant impact of ambient environmental conditions on temperature variation in the NIC canteen. The analysis of the study shows that relative humidity, UV intensity and light level have an effect on temperature and the integration of an IoT based data collection model combined with a multiple linear regression analysis provided many insights on the relevant relationships (mentioned in 5.3 below).

The comparatively high correlation of temperature and humidity stays consistent with existing literature on thermal comfort control (Mutombo & Numbi 2022), which point to the role of humidity in maintaining a stable indoor climate.

5.2 Recommendations

5.2.1 Improving the model

The MLR model can be scaled to fit a larger sample by using back substitution of data and a real time IoT setup to feed in data constantly.

Since the analysis was performed for early year conditions of mainly February and March, it can be scaled to fit a yearly dataset and advanced time series or clustering models can be implemented.

Other environmental factors such as wind speed or CO_2 levels may effect the temperature as well which would increase the accuracy of the model if considered.

5.2.2. Implementation

The sensors will be set up in the canteen area collecting data periodically to predict the temperature, and measures of thermal control can be performed accordingly.

Since the canteen area is open, automated HVAC systems such as air conditioning cannot be used. But automated mechanical fans can be considered during peak humidity.

Extending the roof overhang or using reflective materials above it for reduced UV exposure can be considered.

5.3 Conclusions

The research problem is addressed successfully by the findings of this analysis. An IoT based data collection system paired with a multiple linear regression model confirms that there are significant relationships between external environmental factors such as relative humidity, UV intensity and light level with temperature given by the model,

$$\hat{Y} = (0.724744) + (-0.888851)X_1 + (-0.002041)X_2 + (0.144616)X_3 \pm (0.0611)$$

With a relatively high accuracy as shown by performance metrics (4.1.3.8 above).

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Attachments

IoT and connection management code

https://github.com/DesanSilva/Temperature-Analysis

Research Papers and Datasheets

https://drive.google.com/drive/folders/13DpScLfbOZ9zRrP_oWdc88Ci1vnCeykj?usp=drive_link